

Lecture 2: Protocol layers, OSI Model TCP/P stack and Introduction to IPV4

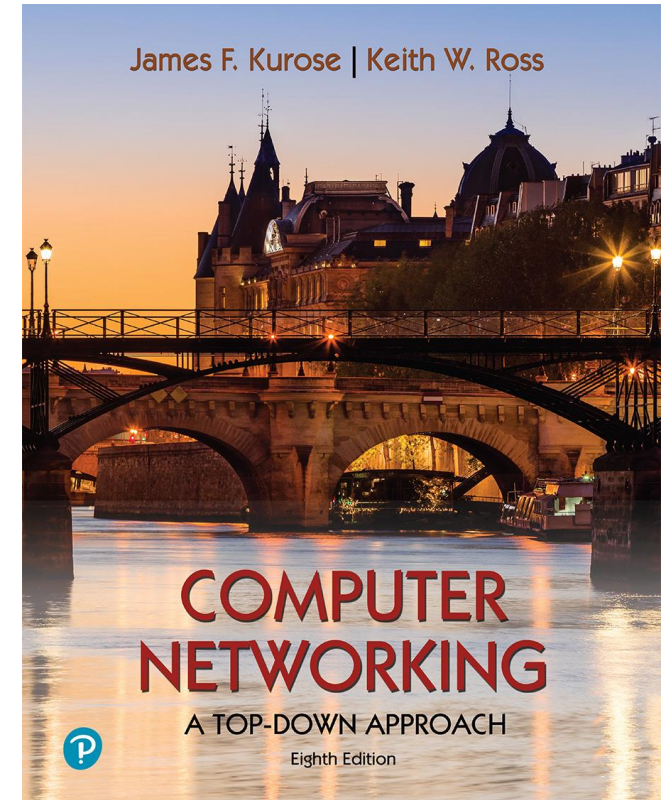
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Protocol “layers” and reference models

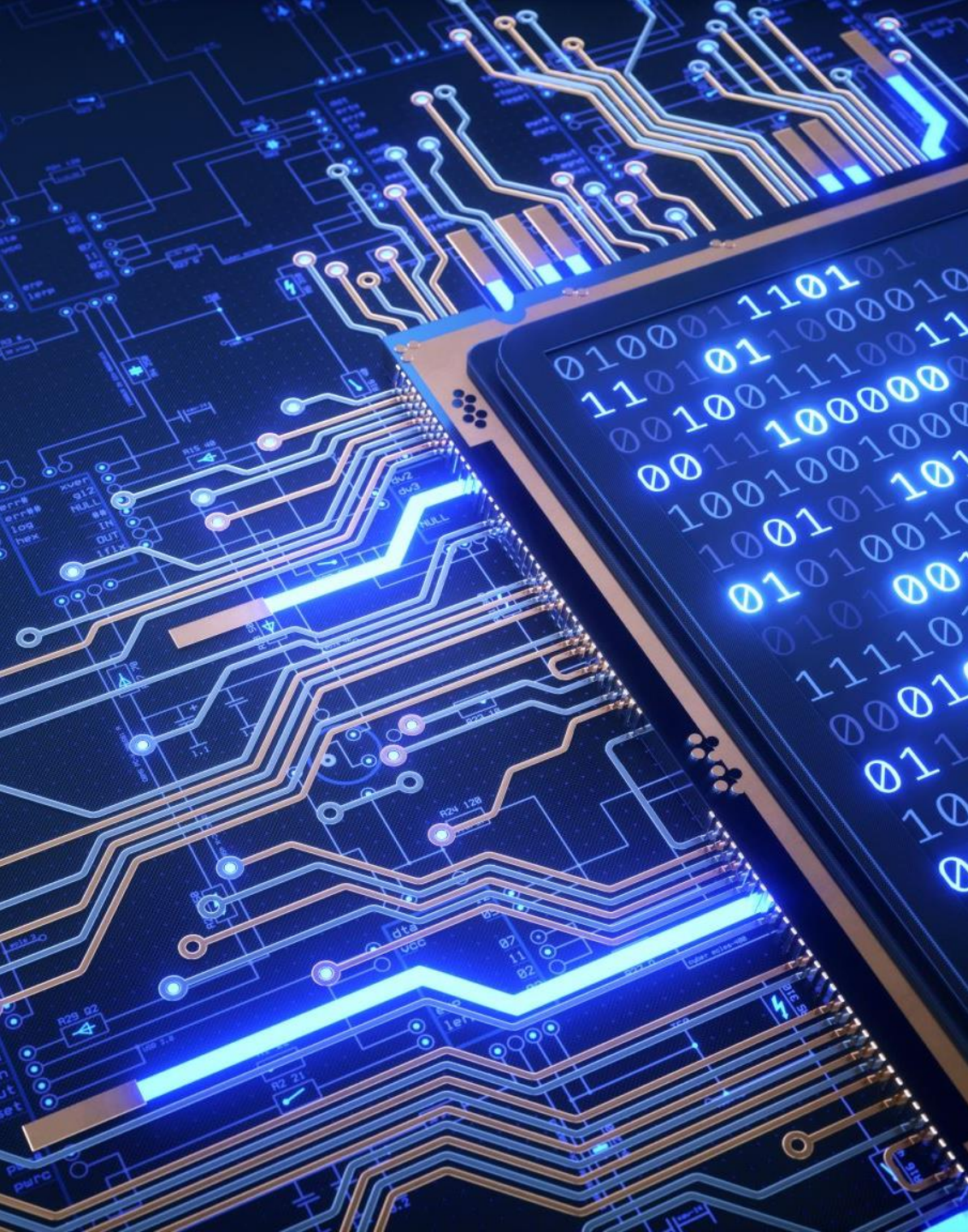
- Slide page
- นำมาจาก Powerpoint slides *Chapter 1: Computer Networking: A Top-Down Approach*
8th edition
โดย Jim Kurose, Keith Ross Source ของ Slides Chapter1:
Introduction page 1-67 – 1-77
 - https://gaia.cs.umass.edu/kurose_ross/ppt.php



Computer Networking: A Top-Down Approach

8th edition

Jim Kurose, Keith Ross
Pearson, 2020



Outcome

- Understand why protocol layer
- Understand OSI Model/ TCP/IP Protocol stacks
- Understand roles and responsibilities of each protocol layer
- Able to give examples of protocols in each layer

Outline

การส่งจดหมาย **Vs** การส่งข้อมูลในระบบเครือข่าย

What is a protocol?

Why Layer?

OSI Model and TCP/IP stack

Service Layer and Encapsulation

Introduction to IPV4

OutCome

- Understand why protocol layer
- Understand OSI Model/ TCP/IP Protocol stacks
- Understand roles and responsibilities of each protocol layer
- Able to give examples of protocols in each layer

การส่งจดหมาย

- คน:
- ข้อความ
- เส้นทางที่ส่ง
- คนที่เกี่ยวข้องในการส่ง
- รหัสไปรษณีย์มีไว้ทำอะไร

Q1: เราจำเป็นต้องรู้หรือไม่ว่าจดหมายเรากว่าจะไปถึงผู้รับผ่านเส้นทางอย่างไร

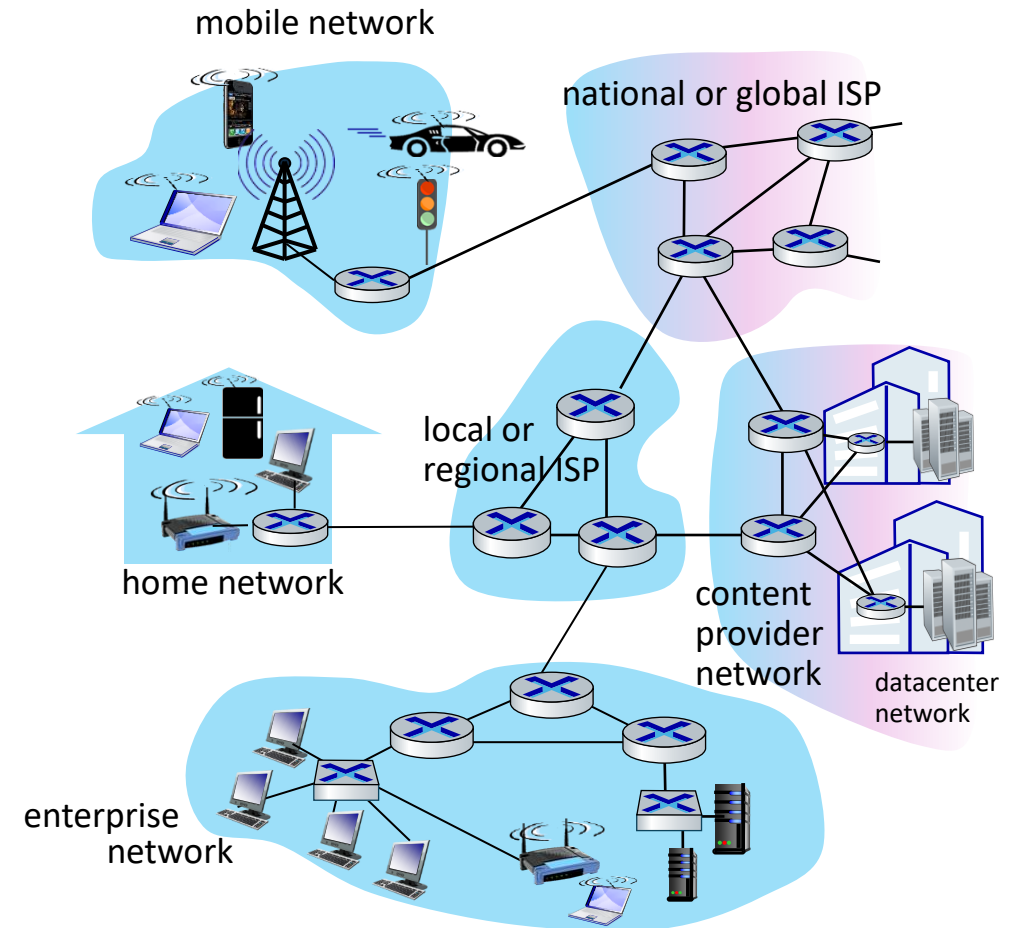
Q2: ในการส่งจดหมาย ถ้าเราต้องการมั่นใจว่าจดหมายที่เราส่งนั้นถึงมือผู้รับแน่นอนต้องทำอะไร

Q3: ถ้าเราส่งจดหมายให้คนต่างชาติที่ไม่ใช่คนไทย ถ้าเราเขียนจดหมายด้วยภาษาไทยชาวต่างชาติจะอ่านออกหรือไม่

3.1 ถ้าอยากให้คนต่างชาติอ่านจดหมายเราได้ เราต้องทำอะไร

การส่งจดหมาย Vs การส่งข้อมูลในระบบเครือข่าย

- คน:
- ข้อความ
- เส้นทางที่ส่ง
- คนที่เกี่ยวข้องในการส่ง



What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

... specific messages sent

... specific actions taken
when message received,
or other events

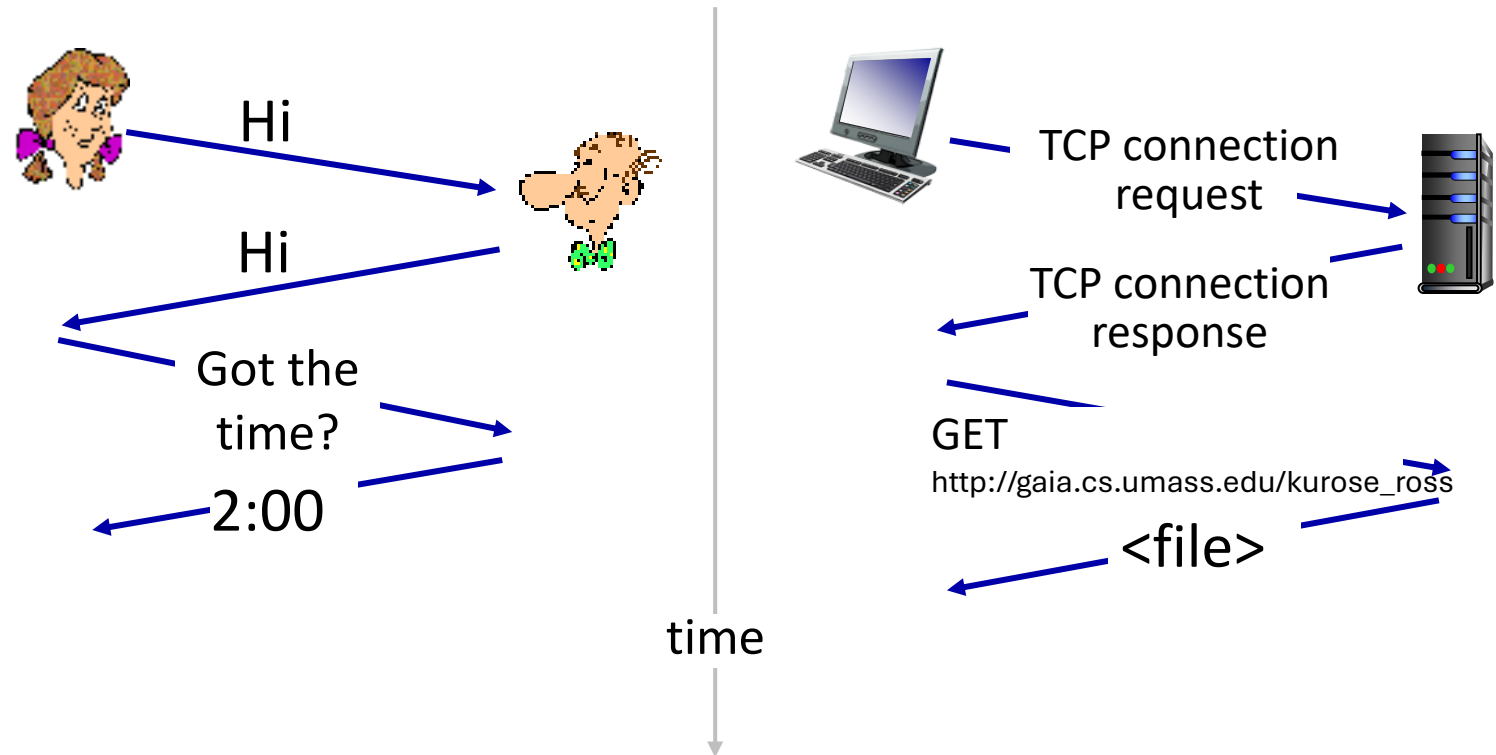
Network protocols:

- computers (devices) rather than humans
- all communication activity in Computer network or the Internet governed by protocols

*Protocols define the **format, order** of
messages sent and received
among network entities, and
actions taken on message
transmission, receipt*

Example Human protocol VS Network protocol

A human protocol and a computer network protocol:



Q: other human protocols?

Protocol “layers” and reference models

Networks are complex,
with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any
hope of *organizing*
structure of network?

- and/or our *discussion*
of networks?

Why layering?

Approach to designing/discussing complex systems:

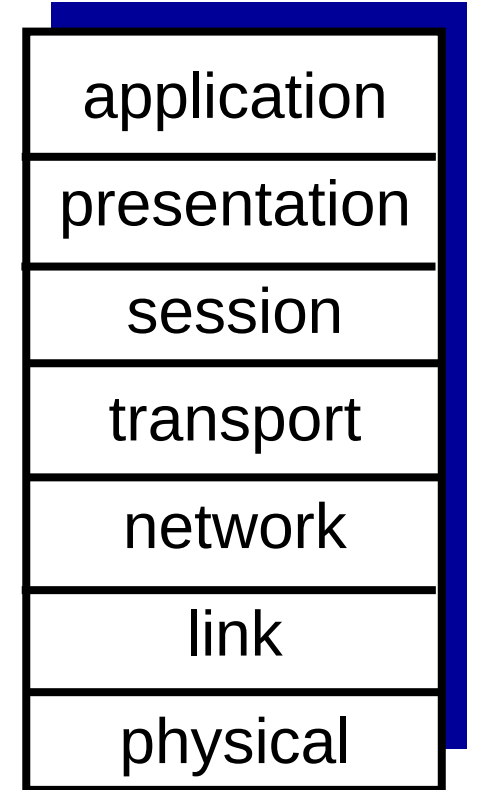
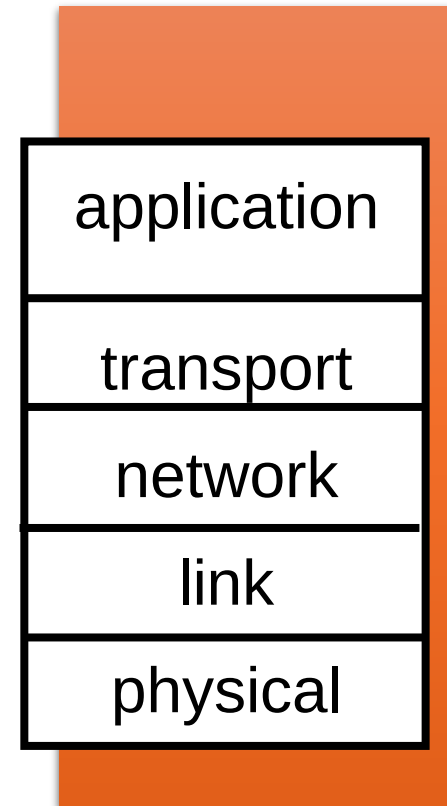
- explicit structure allows identification, relationship of system's pieces
 - layered *reference model* for discussion
- modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system

2 reference models : OSI and TCP/IP Stack

Both implements *layers*:

each layer implements a service

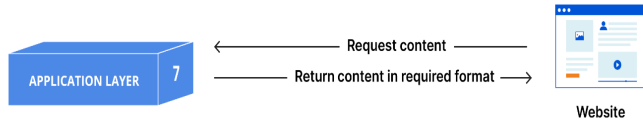
- via its own internal-layer actions
- relying on services provided by layer below
- 2 reference models:
 - OSI model : 7 layer model
 - Internet Protocol Stack (TCP/IP stack)



The seven layer OSI/ISO reference model

OSI Model

Application Layer



The Presentation Layer



The Session Layer



Transport Layer



APPLICATION LAYER

7

— Human-computer interaction layer, where applications can access the network services

PRESENTATION LAYER

6

— Ensures that data is in a usable format and is where data encryption occurs

SESSION LAYER

5

— Maintains connections and is responsible for controlling ports and sessions

TRANSPORT LAYER

4

— Transmits data using transmission protocols including TCP and UDP

NETWORK LAYER

3

— Decides which physical path the data will take

DATA LINK LAYER

2

— Defines the format of data on the network

PHYSICAL LAYER

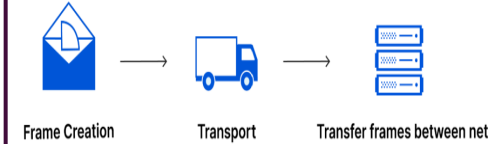
1

— Transmits raw bit stream over the physical medium

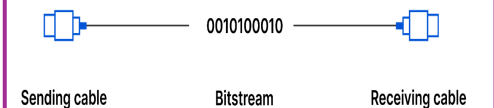
The Network Layer



The Data Link Layer



The Physical Layer



Application Layer

- only layer that directly interacts with data from the user.
- Software applications like web browsers and email clients rely on the application layer to initiate communications with servers.
- the application layer is responsible for the protocols and data manipulation that the software relies on to present meaningful data to the user.
- Application layer protocols include [HTTP](#) as well as [SMTP](#) (Simple Mail Transfer Protocol is one of the protocols that enables [email](#) communications).

Presentation layer

- responsible for preparing data for the application layer;
 - Translation (encoding) , [encryption](#), and compression of data.
- Two communicating devices communicating may be using different encoding methods, so layer 6 is responsible for translating incoming data into a syntax that the application layer of the receiving device can understand.
- If the devices are communicating over an encrypted connection, layer 6 is responsible for adding the encryption on the sender's end as well as decoding the encryption on the receiver's end so that it can present the application layer with unencrypted, readable data.
- Finally the presentation layer is also responsible for compressing data it receives from the application layer before delivering it to layer 5. This helps improve the speed and efficiency of communication by minimizing the amount of data that will be transferred.

Session layer

- This layer is responsible for opening and closing communication between the two devices.
- The time between when the communication is opened and closed is known as the session.
- The session layer ensures that
 - the session stays open long enough to transfer all the data being exchanged, and then
 - promptly closes the session in order to avoid wasting resources.
- The session layer also synchronizes data transfer with checkpoints. For example, if a 100 megabyte file is being transferred, the session layer could set a checkpoint every 5 megabytes. In the case of a disconnect or a crash after 52 megabytes have been transferred, the session could be resumed from the last checkpoint, meaning only 50 more megabytes of data need to be transferred. Without the checkpoints, the entire transfer would have to begin again from scratch.

Transport layer

Layer 4

- taking data from the session layer and breaking it up into chunks called **segments** before sending it to layer 3.
- The transport layer on the receiving device is responsible for **reassembling the segments into data** the session layer can consume.

- Layer 4 is responsible for
 - **end-to-end communication** between the **two devices**.
 - process-process data transfer
- The transport layer is also responsible for **flow control** and **error control**.
 - Flow control determines an optimal speed of transmission to ensure that a sender with a fast connection does not overwhelm a receiver with a slow connection.
 - The transport layer performs error control on the receiving end by ensuring that the data received is complete, and requesting a retransmission if it isn't.

Network Layer

- The network layer
 - breaks up segments from the transport layer into smaller units, called packets, on the sender's device, and
 - reassembling these packets on the receiving device.

- The network layer is responsible
 - for facilitating data transfer **between two different networks.**
 - If the two devices communicating are on the same network, then the network layer is unnecessary.
- The network layer also **finds the best physical path for the data to reach its destination;** this is known as routing.
- Network layer protocols include IP, the Internet Control Message Protocol (ICMP), the Internet Group Message Protocol (IGMP), and the IPsec suite.

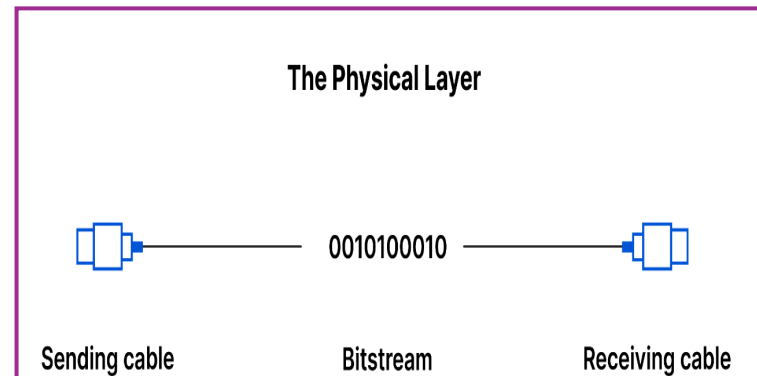
Data Link Layer

- The data link layer takes packets from the network layer and breaks them into smaller pieces called frames.

- The data link layer is very similar to the network layer, except
 - the data link layer facilitates data transfer between two devices on the *same* network.
- the data link layer is also responsible for flow control and error control in intra-network communication (The transport layer only does flow control and error control for inter-network communications).

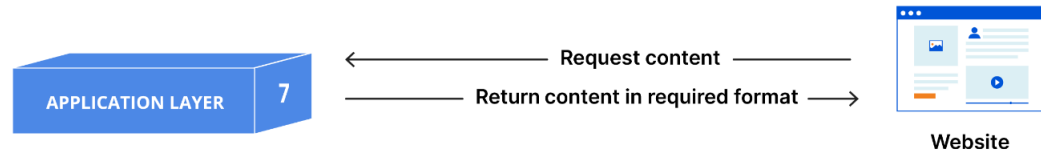
Physical layer

- the layer
 - where the **data** gets converted into a **bit stream**, which is a string of 1s and 0s.
 - includes the physical equipment involved in the data transfer, such as the cables and switches.
- The physical layer of both devices must also agree on a signal convention so that the 1s can be distinguished from the 0s on both devices.

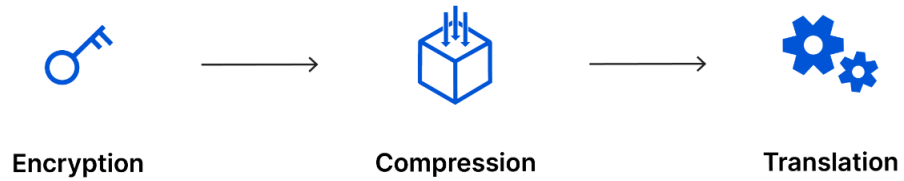


OSI Model

Application Layer



The Presentation Layer



The Session Layer



<https://www.cloudflare.com/learning/ddos/glossary/open-systems-interconnection-model-osi/>

Transport Layer



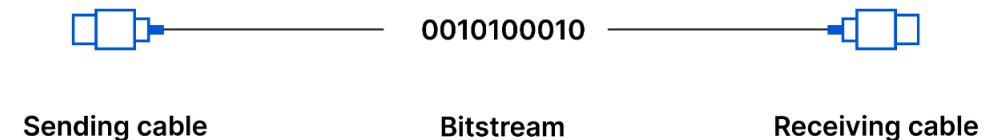
The Network Layer



The Data Link Layer

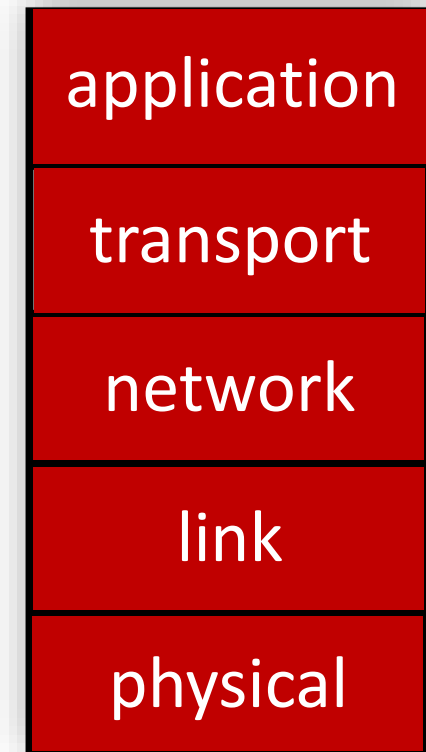


The Physical Layer



Layered Internet protocol stack

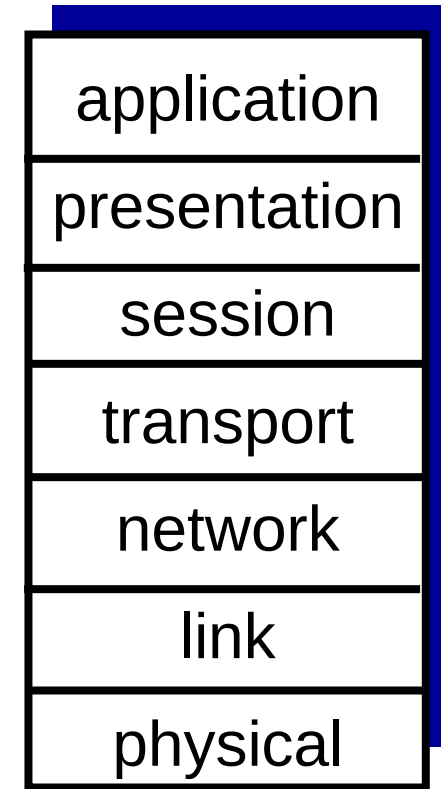
- *application*: supporting network applications
 - HTTP, IMAP, SMTP, DNS
- *transport*: process-process data transfer
 - TCP, UDP
- *network*: routing of datagrams from source to destination
 - IP, routing protocols
- *link*: data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- *physical*: bits “on the wire”



ISO/OSI reference model

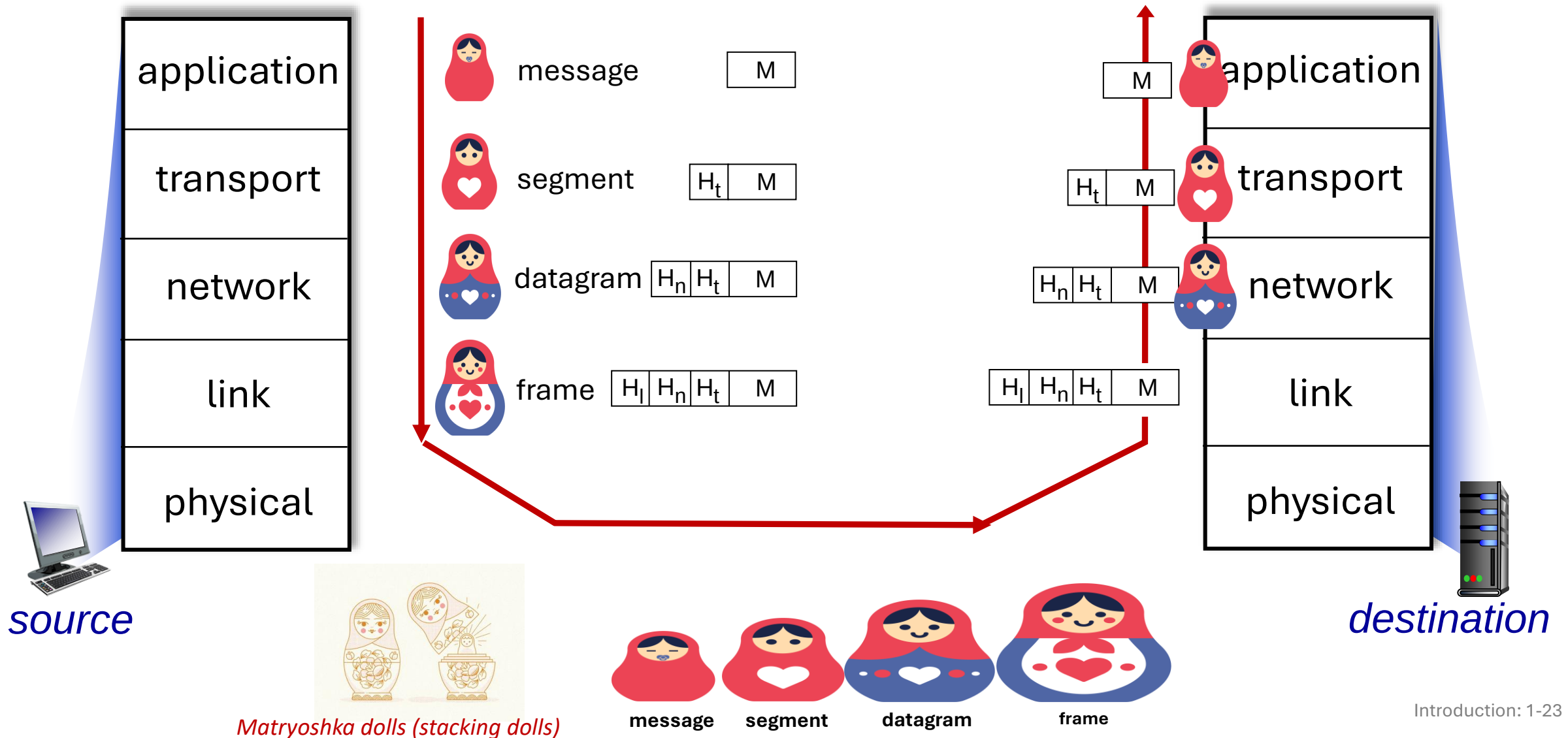
Two layers not found in Internet protocol stack!

- *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- *session*: synchronization, checkpointing, recovery of data exchange
- Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application

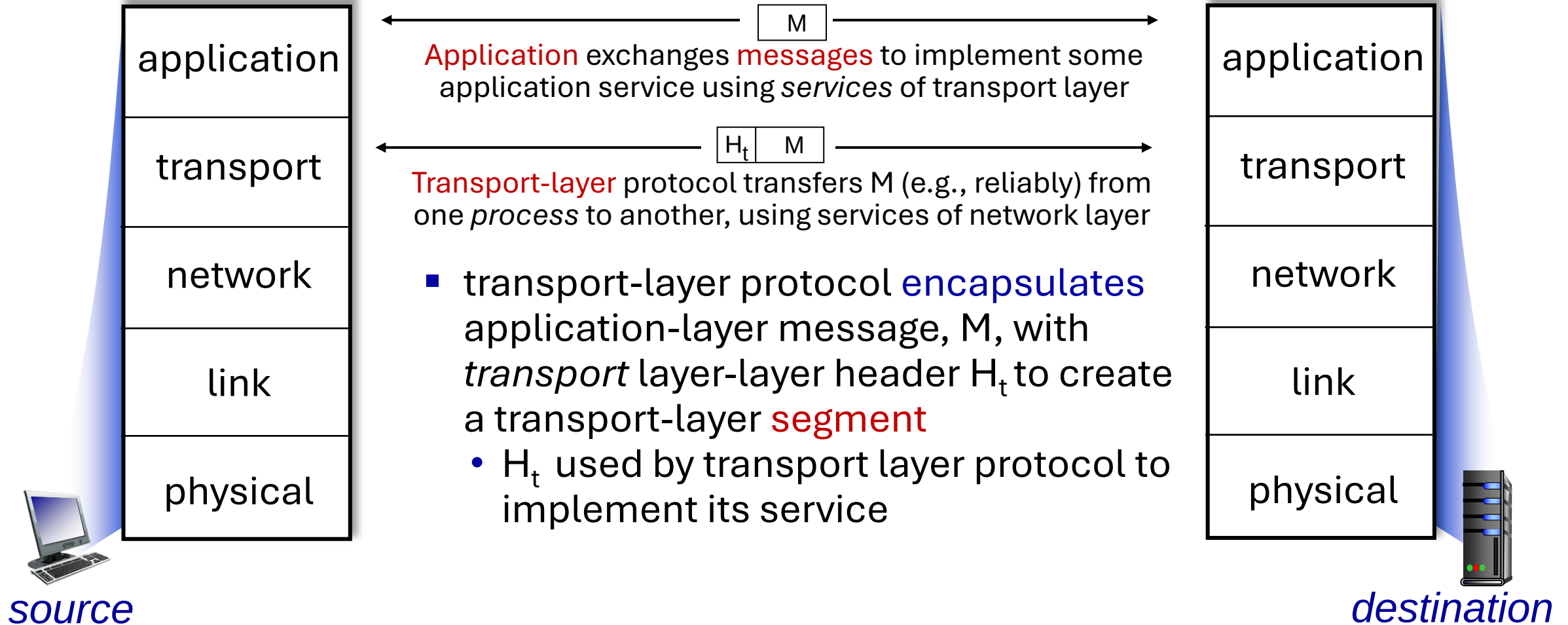


The seven layer OSI/ISO reference model

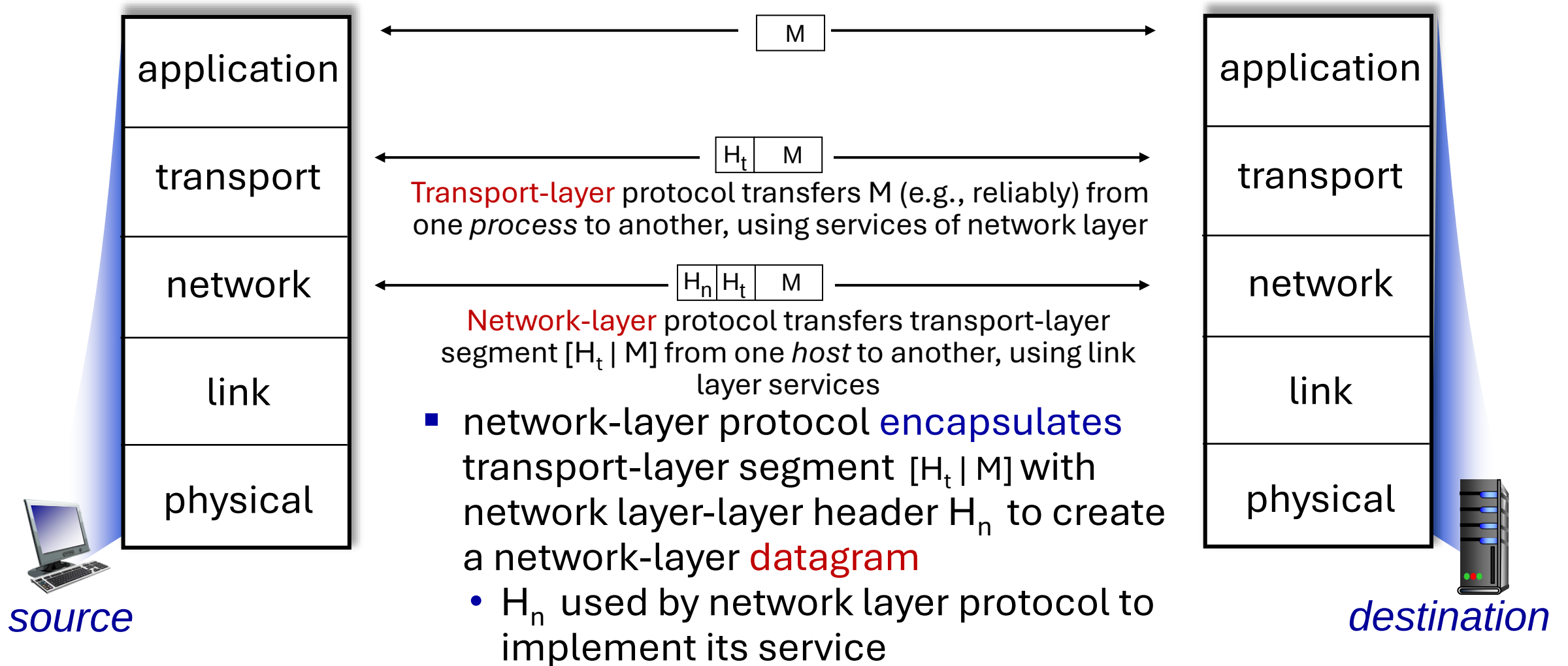
Services, Layering and Encapsulation



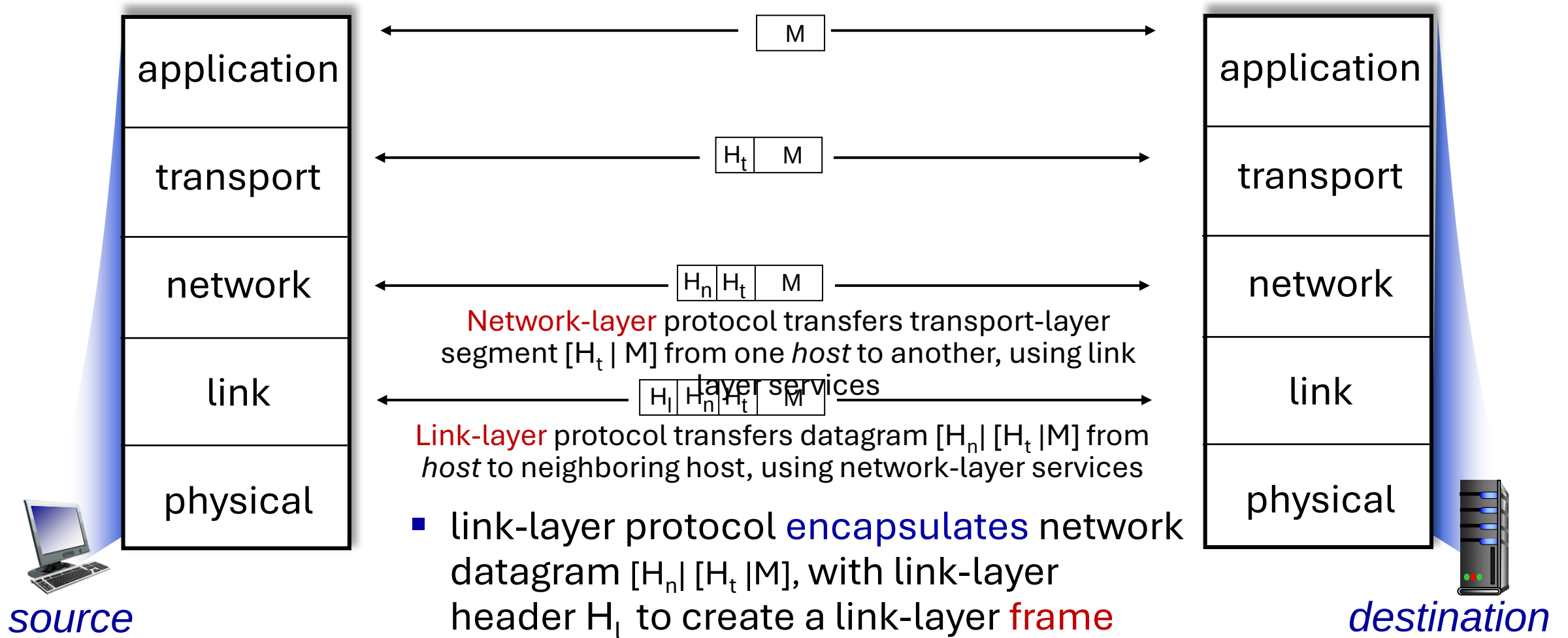
Services, Layering and Encapsulation



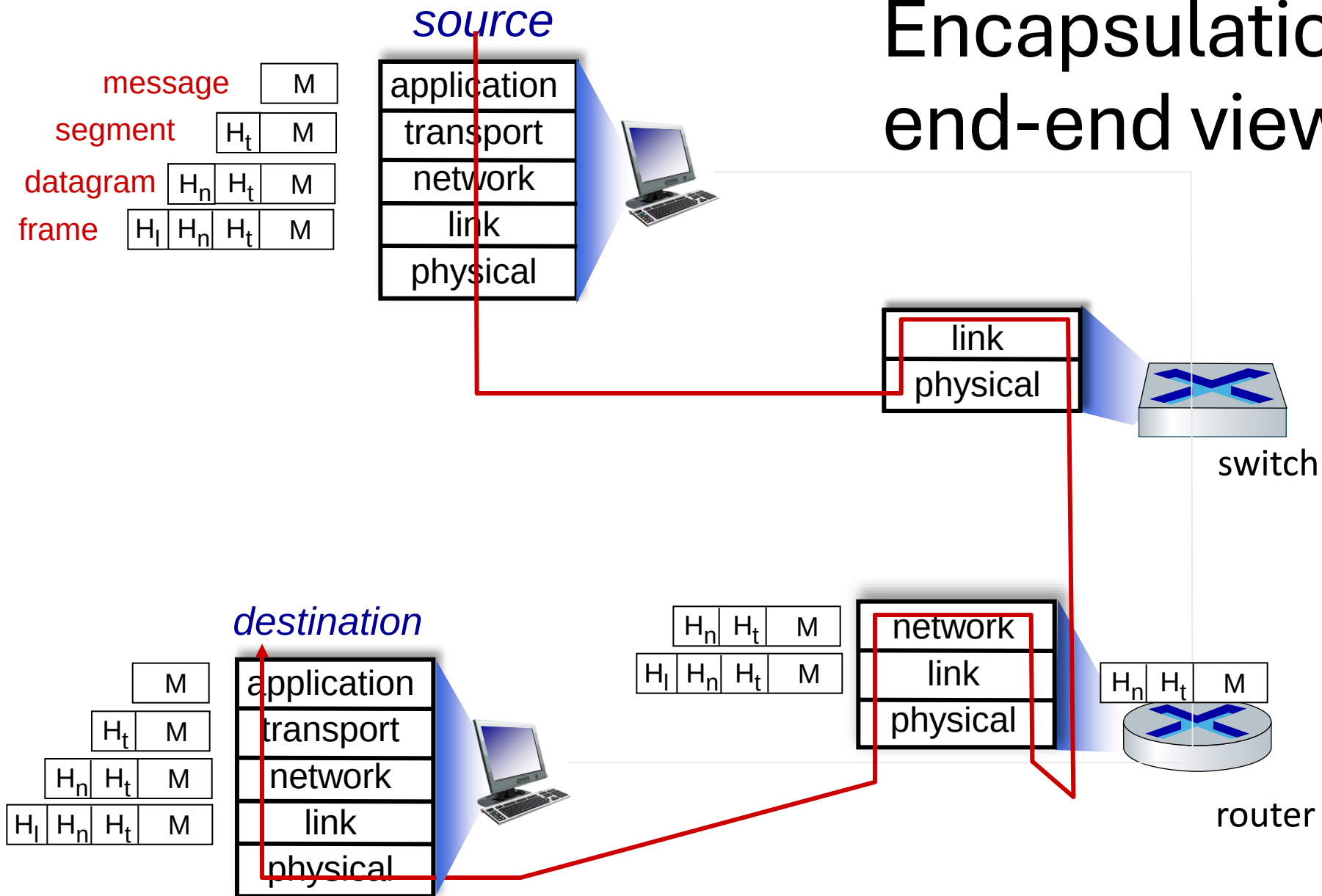
Services, Layering and Encapsulation



Services, Layering and Encapsulation



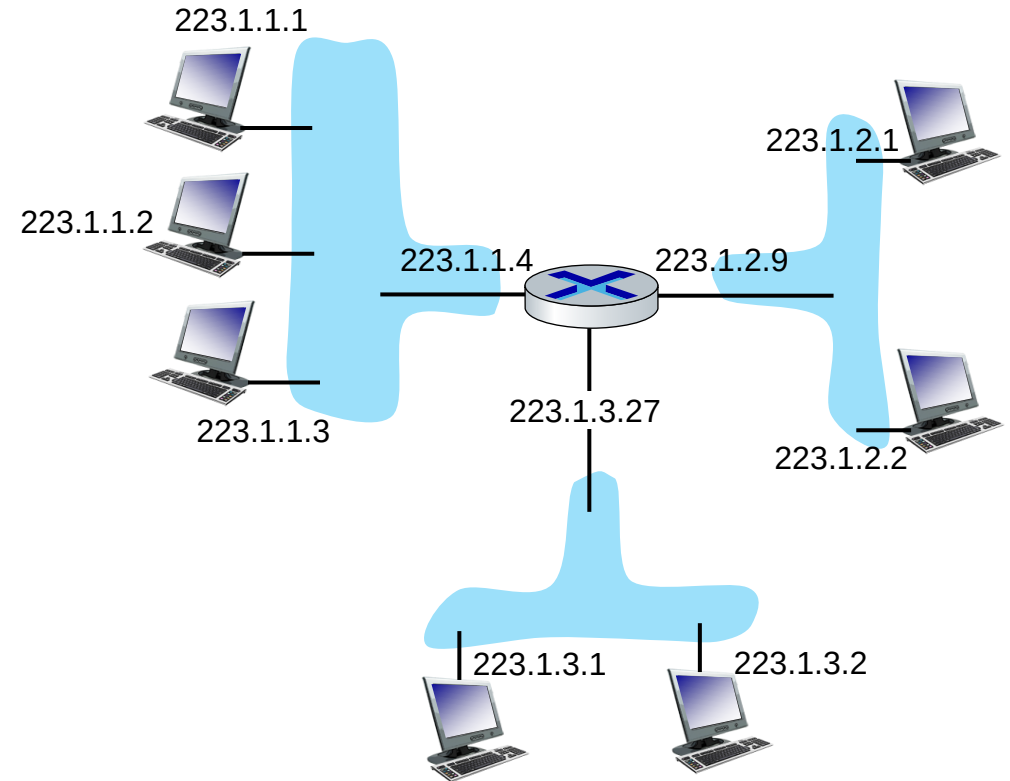
Encapsulation: an end-end view



Introduction to IP address

IP addressing: introduction

- **IP address v4**: 32-bit identifier associated with each host or router *interface*
- **interface**: connection between host/router and physical link
 - router's typically have **multiple interfaces**
 - host typically **has one or two interfaces** (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

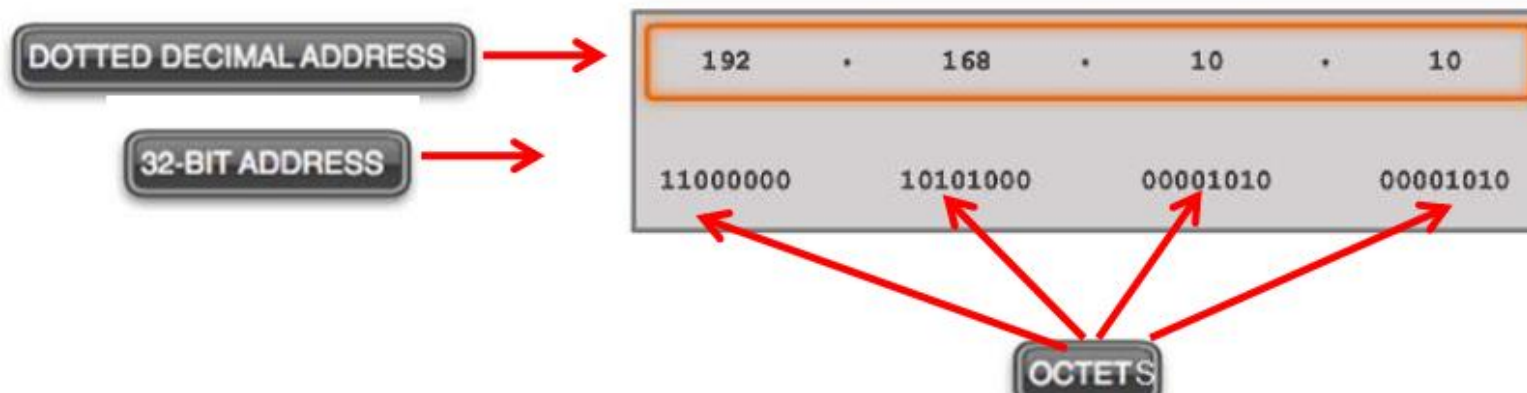
223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1



IPv4 Address Structure

Binary Number System



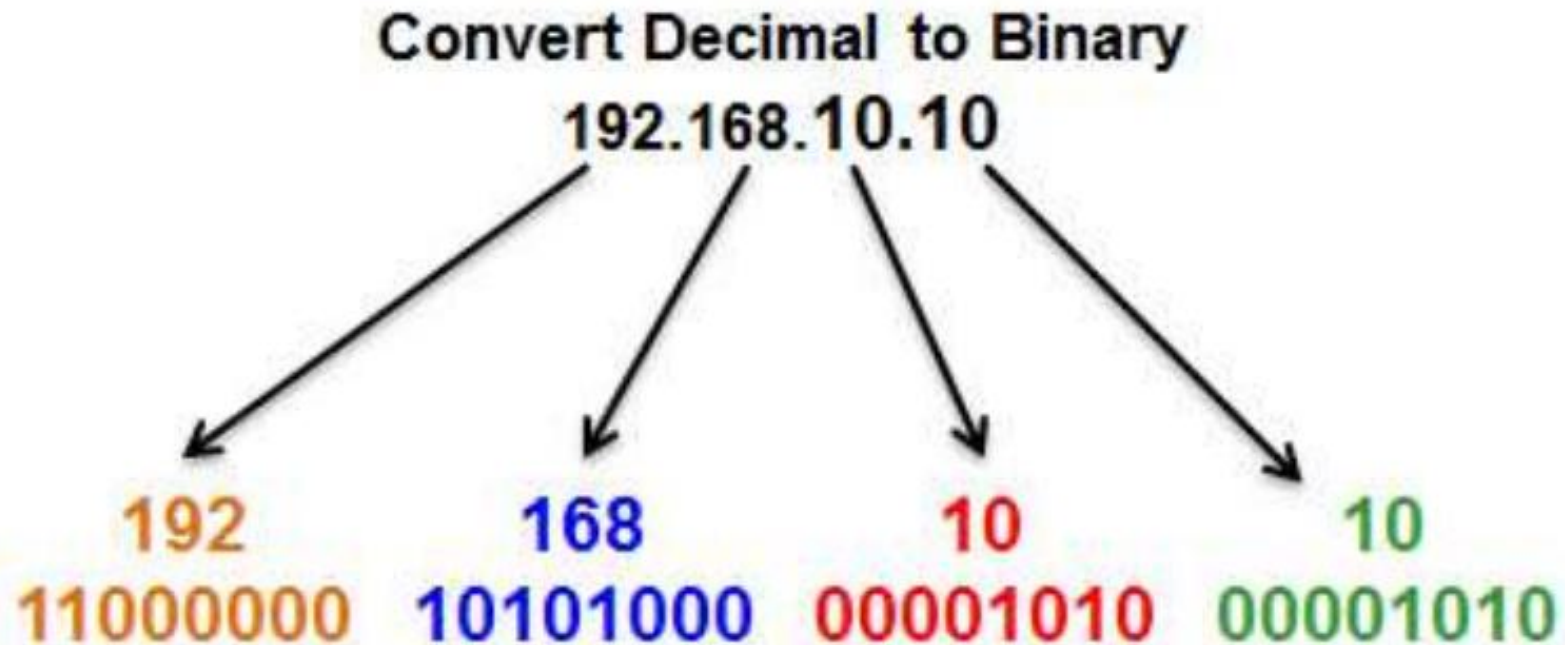
Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Octet Bit Values	128	64	32	16	8	4	2	1
Binary Address	1	1	0	0	0	0	0	0
Binary Bit Values	128	64	0	0	0	0	0	0

Add the binary bit values.
 $128 + 64 = 192$



IPv4 Address Structure

Converting from Decimal to Binary Conversions



Exercise 1:

Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Octet Bit Values	128	64	32	16	8	4	2	1
Binary Address	1	1	0	0	0	0	0	0
Binary Bit Values	128	64	0	0	0	0	0	0

- เปลี่ยนเลขฐานสองของ IP address ต่อไปนี้เป็นหมายเลข IP ที่แทนด้วยเลขฐานสิบ (มี . ด้วยค่ะ)
 - 10000001 00001011 00001011 11101111

Exercise 2:

- เปลี่ยนหมายเลข IP ต่อไปนี้เป็น เลขฐานสอง 32 บิต
 - 111.56.45.78

Radix	2	2	2	2	2	2	2	2
Exponent	7	6	5	4	3	2	1	0
Octet Bit Values	128	64	32	16	8	4	2	1
Binary Address	1	1	0	0	0	0	0	0
Binary Bit Values	128	64	0	0	0	0	0	0

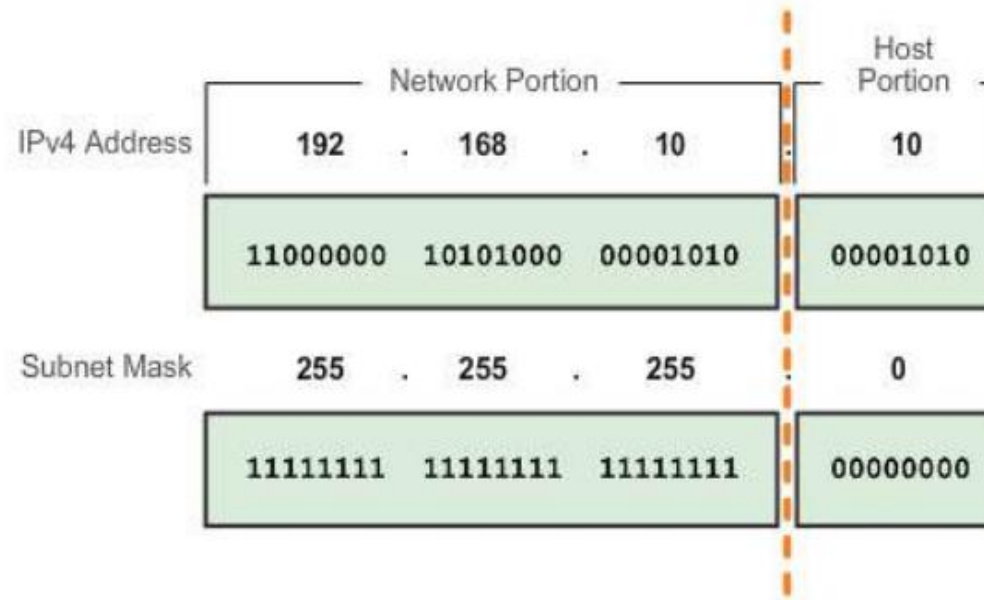
Exercise 3:

- ทำไมหมายเลขเหล่านี้ไม่ใช่ Valid IP address
 - 111.56.045.78
 - 75.45.301.14



IPv4 Subnet Mask

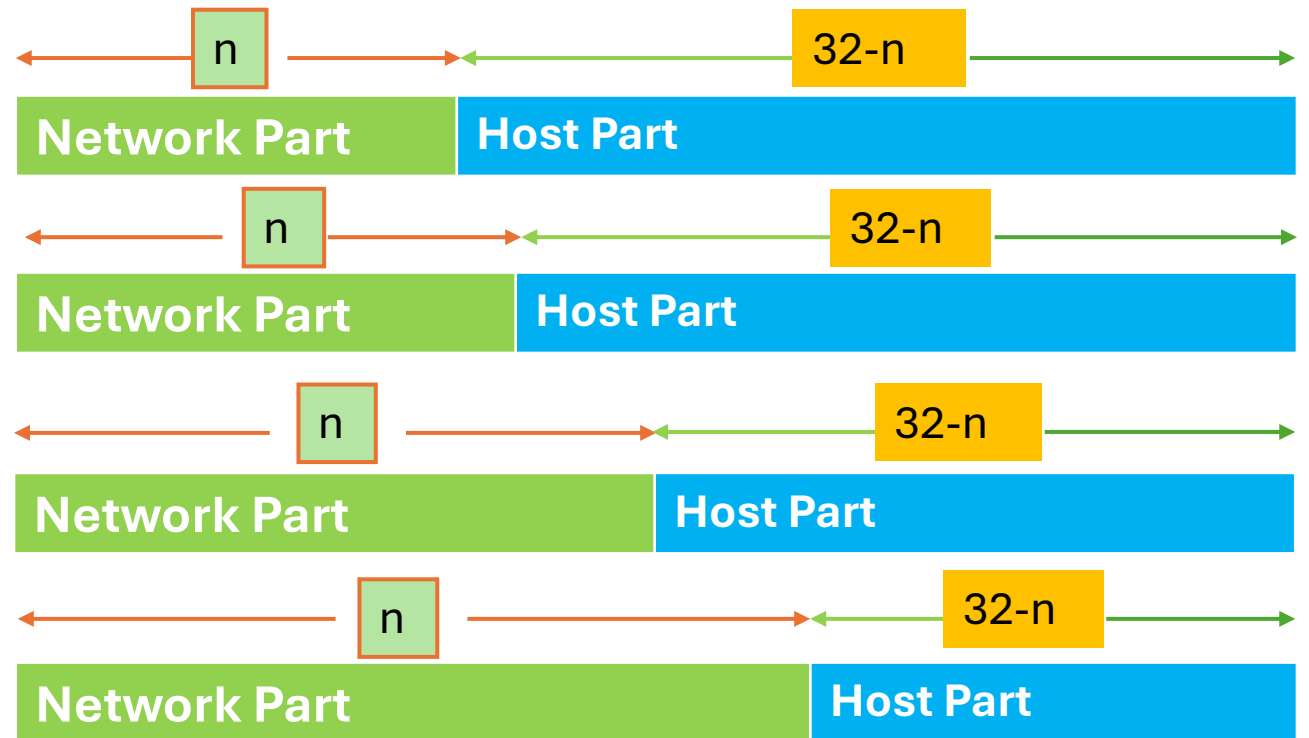
Network Portion and Host Portion of an IPv4 Address



- To define the network and host portions of an address, a device uses a separate 32-bit pattern called a subnet mask
- The subnet mask does not actually contain the network or host portion of an IPv4 address, it just says where to look for these portions in a given IPv4 address

IPv4 Component

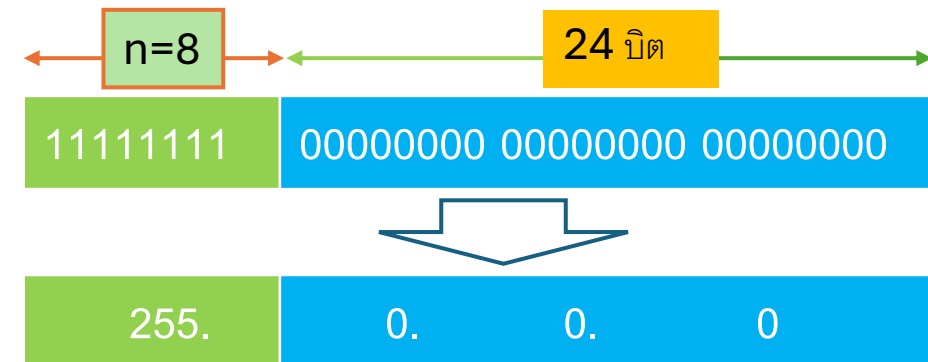
- แบ่งเป็น 2 ส่วน
 - Network Part
 - N bits
 - Host Part
 - 32-N bits
- ถ้า network part n บิต
บางครั้งจะเขียนว่า /n



IP Subnet Mask

- IP Subnet mask **Separate** ระหว่าง **Network Part** กับ **Host Part**
 - โดยใน 32 bit ค่าของ subnet mask จะให้ส่วนที่เป็น network part เป็น 1 ทั้งหมดทุกบิต และ ส่วนที่เป็น Host part เป็น 0 ทุกบิต
- IP address สองค่าใดๆ ที่ เมื่อนำค่าไป AND กับ subnet mask แล้ว ได้ค่าที่เหมือนกันแสดงว่า IP address สองค่านั้นอยู่ใน network เดียวกัน
- จงหาค่าเลขฐานสิบของ IP Subnet Mask ต่อไปนี้
 - 11111111 11111111 00000000 00000000
 - 11111111 11111111 11111111 00000000

Subnet Value	Bit Value							
	128	64	32	16	8	4	2	1
255	1	1	1	1	1	1	1	1
254	1	1	1	1	1	1	1	0
252	1	1	1	1	1	1	0	0
248	1	1	1	1	1	0	0	0
240	1	1	1	1	0	0	0	0
224	1	1	1	0	0	0	0	0
192	1	1	0	0	0	0	0	0
128	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0



IP Subnet Mask

(Exercise 4)

11111111 00000000 00000000 00000000

255. 0. 0. 0

11111111 11111111 00000000 00000000

? . ? . ? . ?

11111111 11111111 11111111 00000000

? . ? . ? . ?

Subnet Value	Bit Value							
	128	64	32	16	8	4	2	1
255	1	1	1	1	1	1	1	1
254	1	1	1	1	1	1	1	0
252	1	1	1	1	1	1	0	0
248	1	1	1	1	1	0	0	0
240	1	1	1	1	0	0	0	0
224	1	1	1	0	0	0	0	0
192	1	1	0	0	0	0	0	0
128	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

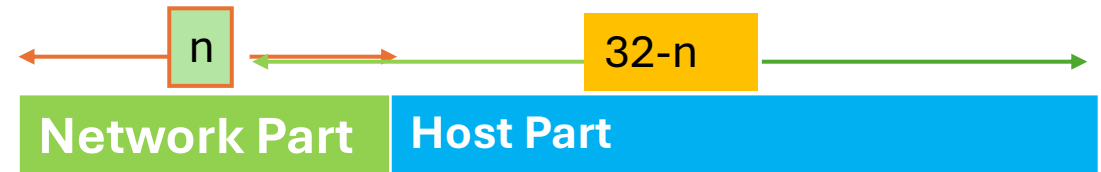
/1	10000000 00000000 00000000 00000000	128.0.0.0
/2		
/3		
/4		
/5		
/6		
/7		
/8		
/9		
/10		
/11		
/12		
/13		
/14		
/15		
/16		
/17		

	Binary IP	Decimal IP
/1	10000000 00000000 00000000 00000000	128.0.0.0
/2		
/3		
/4		
/5		
/6		
/7		
/8		
/9		
/10		
/11		
/12		
/13		
/14		
/15		
/16		
/17		

	Binary IP	Decimal IP
/18	10000000 00000000 00000000 00000000	128.0.0.0
/19		
/20		
/21		
/22		
/23		
/24		
/25		
/26		
/27		
/28		
/29		
/30		
/31		
/32		

How about Host part

- **Host part** คือ ค่าที่กำหนดให้กับแต่ละ **Host** ซึ่งแต่ละค่าจะไม่เหมือนกัน
- จำนวนบิตของ **Host part** จะทำให้สามารถคำนวณได้ว่า ใน **Lan** วงนั้นจะสามารถมีหมายเลข **IP address** ได้ทั้งหมดกี่หมายเลข
- จากจำนวน **IP** ที่คำนวณได้จาก **Host part** จะสามารถทราบ
 - จำนวน **IP** ที่สามารถ **assign** ให้ **Host**
 - หมายเลข **IP** ที่สามารถ **assign** ให้ **Host**



- การคำนวณ
 - Network part n bit
 - ดังนั้น $\text{host part} = 32 - n$ บิต
 - จำนวน **IP** ที่เป็นได้ทั้งหมด
 - $2^{(32-N)}$
- จำนวน **IP** ที่สามารถใช้ได้ ใน **network** นั้น
 - $2^{(32-N)} - 2$
 - -2 คือ
 - 1 IP สำหรับ **network address** และ
 - 1 IP สำหรับ **Broadcast address**

ประเภทของ IP address

Unicast Address

- IP ที่ใช้ระบุแต่ละ Host
- 1 IP ต่อ 1 Host ห้ามซ้ำ

Network address

- IP address ที่ใช้ระบุวงแลนของเครื่องคอมพิวเตอร์ที่สังกัด
- ถ้าพบว่า network address เดียวกันจะอยู่วงแลนเดียวกัน
- บิตส่วนของ Host part จะเป็น 0 ทั้งหมด

Broadcast address

- ในกรณีที่ต้องการส่งข้อมูลหาเครื่องทุกเครื่องที่อยู่วงแลน ไม่จำเป็นต้อง generate 1 packet ต่อ 1 host ซึ่งถ้าทั้งวงแลนมี 10 host ต้อง generate 10 packet
- แต่เราสามารถให้ IP ของ Broadcast address ในการระบุ เป็น Destination IP address แล้ว generate เพียง packet เดียวได้เลย
- ทุก host ในวงแลนเมื่อเห็น packet นี้จะเข้าใจว่าตนต้องรับ packet นี้ไปทำงาน
- บิตส่วนของ Host part จะเป็น 1 ทั้งหมด

Multicast (out of scope here)

	Subnet mask	จำนวน IP ที่ เป็นไปได้ทั้งหมด	จำนวน IP ที่สามารถ Assigned ให้ Host
/1	128.0.0.0		
/2			
/3			
/4			
/5			
/6			
/7			
/8	255.0.0.0		
/9			
/10			
/11			
/12			
/13			
/14			
/15			
/16	255.255.0.0		
/17			

	Subnet mask	จำนวน IP ที่ เป็นไปได้ทั้งหมด	จำนวน IP ที่สามารถ Assigned ให้ Host
/18	128.0.0.0		
/19			
/20			
/21			
/22			
/23			
/24	255.255.255.0		
/25			
/26			
/27			
/28			
/29			
/30			
/31			
/32			

IP addresses: how to get one?

That's actually **two** questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)?

How does *host* get IP address?

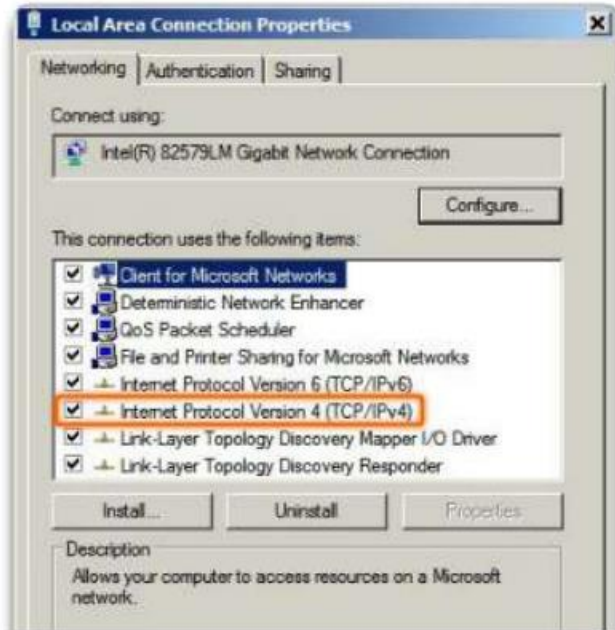
- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- **DHCP**: Dynamic Host Configuration Protocol: dynamically get address from a server
 - “plug-and-play”



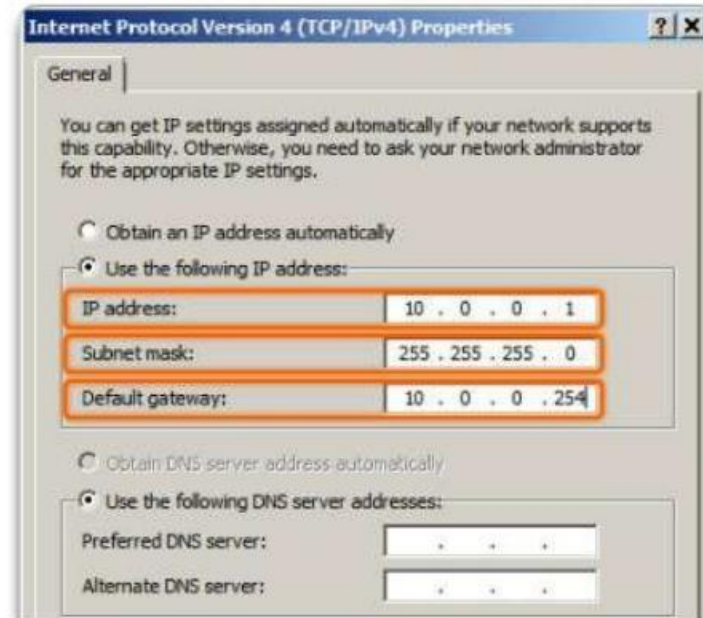
IPv4 Unicast, Broadcast, and Multicast

Assigning a Static IPv4 Address to a Host

LAN Interface Properties



Configuring a Static IPv4 Address



Edit network IP settings

Manual

IPv4

☒ On

IP address

Subnet mask

Gateway

Preferred DNS

DNS over HTTPS

Off

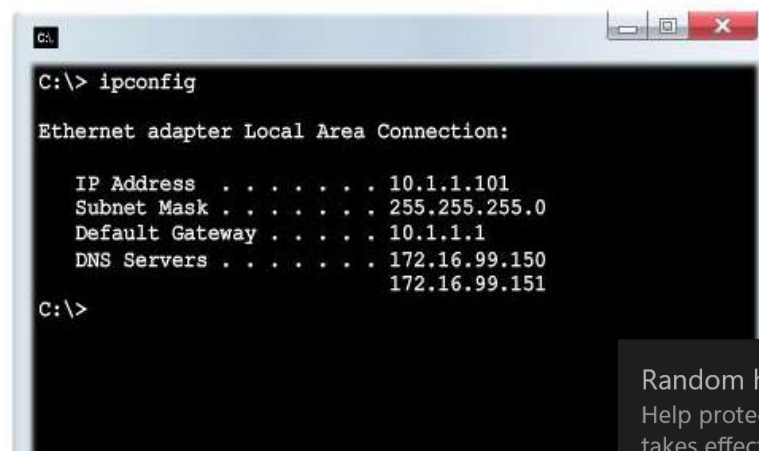
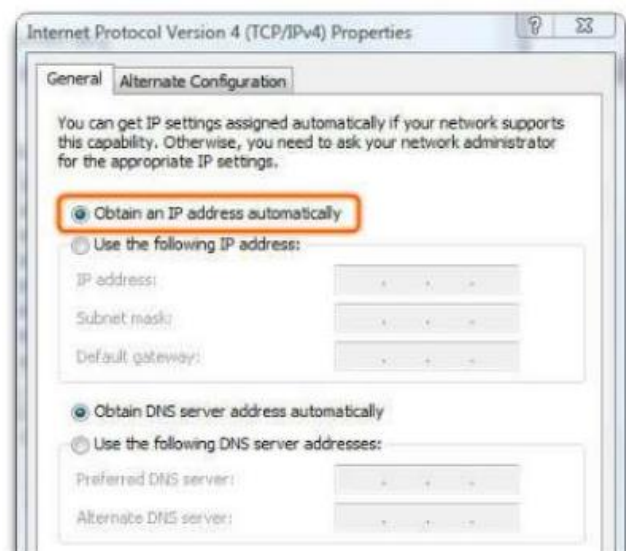
Alternate DNS

Save

Cancel

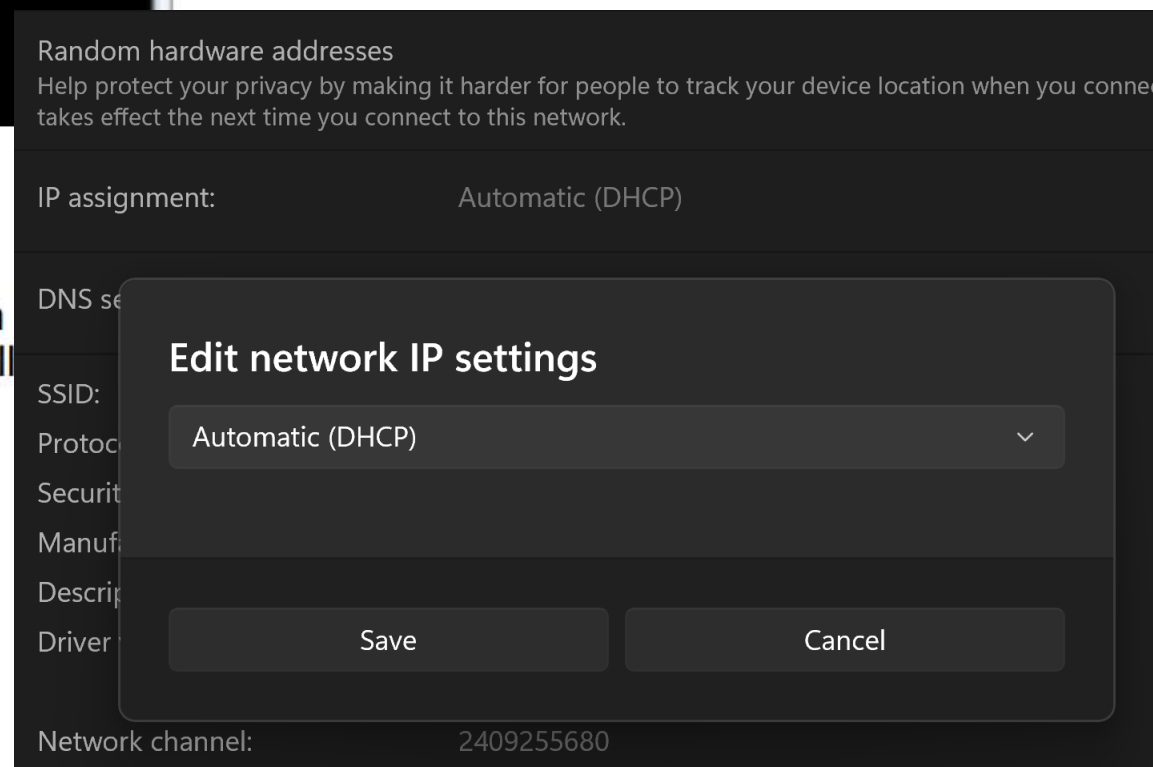
IPv4 Unicast, Broadcast, and Multicast

Assigning a Dynamic IPv4 Address to a Host



Verification

DHCP - preferred method of “leasing” IPv4 addresses to hosts on networks, reduces the burden on network support staff and virtually eliminates entry errors



How to find Network Address

Network Address

=

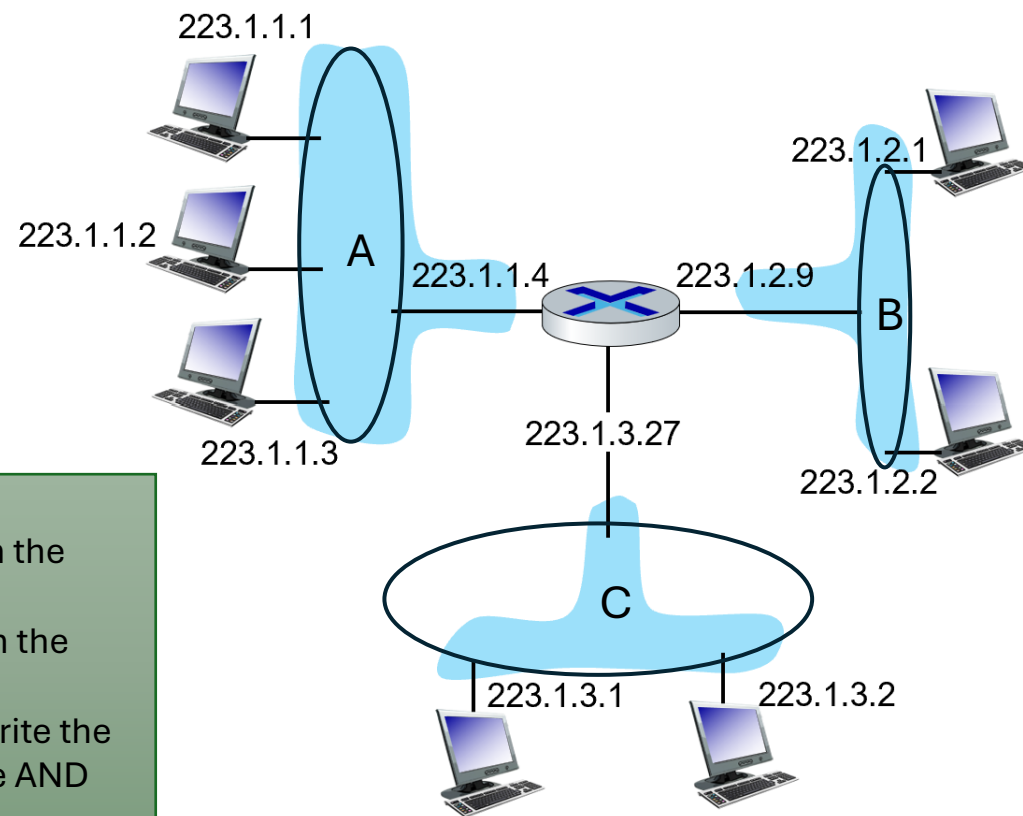
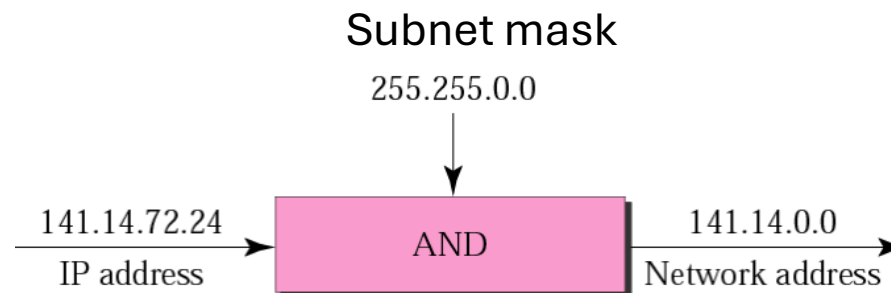
IP Address AND Subnet mask

- จากรูป ถ้าให้ทุกวงแลน มี subnet mask เป็น 255.255.255.0
 - จงหา Network address ของแต่ละ network
- ตัวอย่าง Network C มี 3 IP นักศึกษาลองนำแต่ละตัว And 255.255.255.0

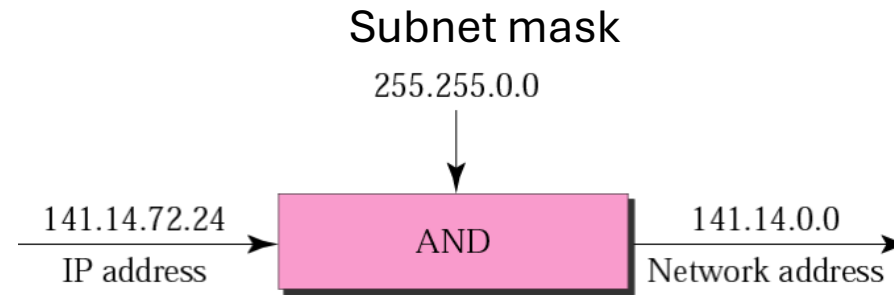
- 223.1.3.1
- 223.1.3.2
- 223.1.3.27

Hints

- If the byte in the mask is **255**, copy the byte in the address.
- If the byte in the mask is **0**, replace the byte in the address with 0.
- If the byte in the mask is **neither 255 nor 0**, write the mask and the address in binary and apply the AND operation.



How to find Network Address (2)



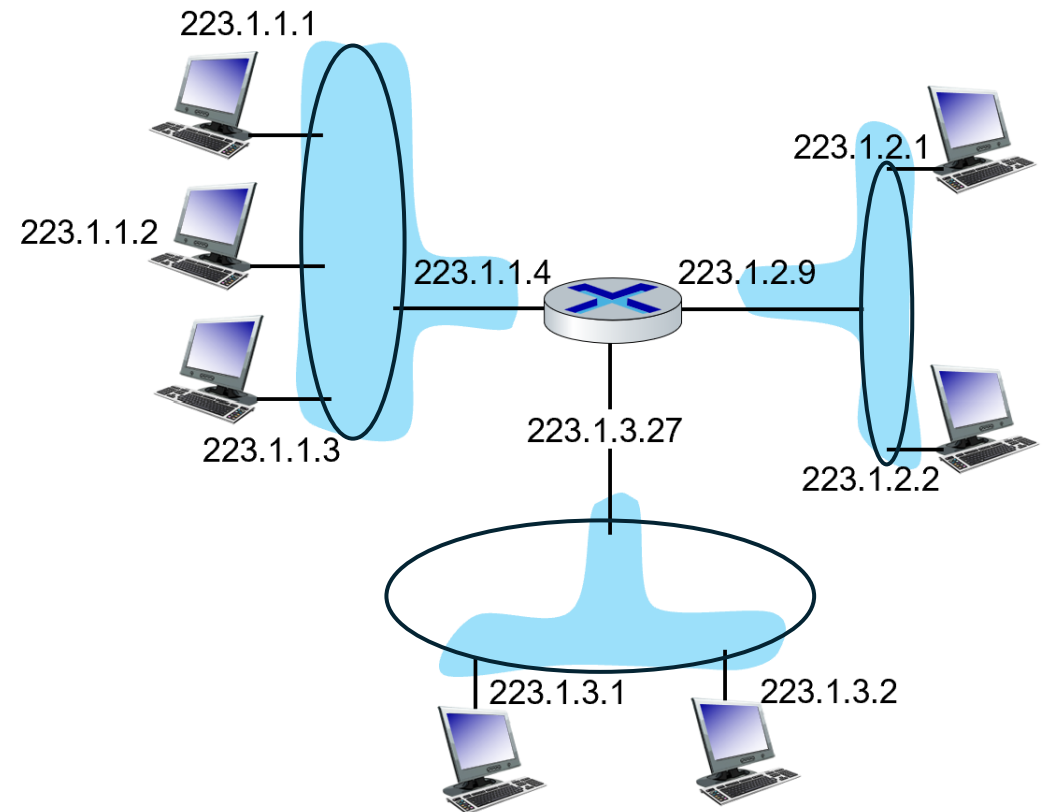
Network Address

=

IP Address **AND** Subnet mask

- จากรูป ถ้าให้ทุกวงแลน มี subnet mask เป็น 255.255.255.0
 - จงหา Network address ของแต่ละ network
- ตัวอย่าง

223	1	3	27
11011111	00000001	00000011	00011011
11111111	11111111	11111111	00000000
11011111	00000001	00000011	00000000
223	1	3	0



How to find Broadcast Address

วิธีคิด Broadcast address แบบที่ 1

- 1. Flip บิตของ subnet สลับ 0 เป็น 1
- 2. นำไป Or กับ IP address

วิธีคิด Broadcast address แบบที่ 2

- 1. Flip บิตของ network address ตรง host part ให้เป็น 1 ทั้งหมด

- Hints
- If the byte in the mask is 255, copy the byte in the address.
 - If the byte in the mask is 0, replace the byte in the address with 0.
 - If the byte in the mask is neither 255 nor 0, write the mask and the address in binary and apply the AND operation.

IP address ของ Host ฐานสิบ	223	1	3	27
IP address ของ Host ฐานสอง	11011111	00000001	00000011	00011011
Subnet (ฐานสอง)	11111111	11111111	11111111	00000000
Network address ฐานสอง	11011111	00000001	00000011	00000000
Network address ฐานสิบ	223	1	3	0
Broadcast address ฐานสอง				
Broadcast address ฐานสิบ				



Lab: Introduction to Wireshark

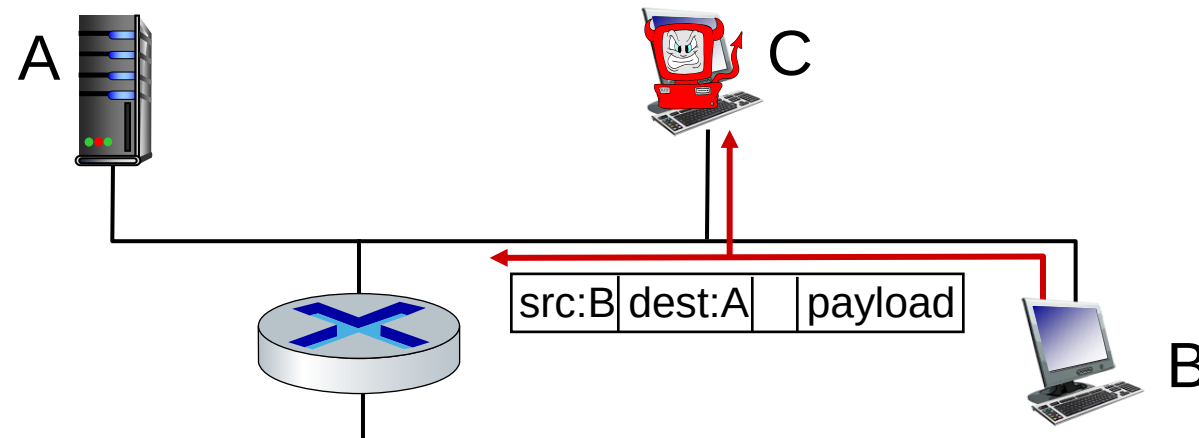
Network security

- Internet not originally designed with (much) security in mind
 - *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!
- We now need to think about:
 - how bad guys can attack computer networks
 - how we can defend networks against attacks
 - how to design architectures that are immune to attacks

Bad guys: packet interception

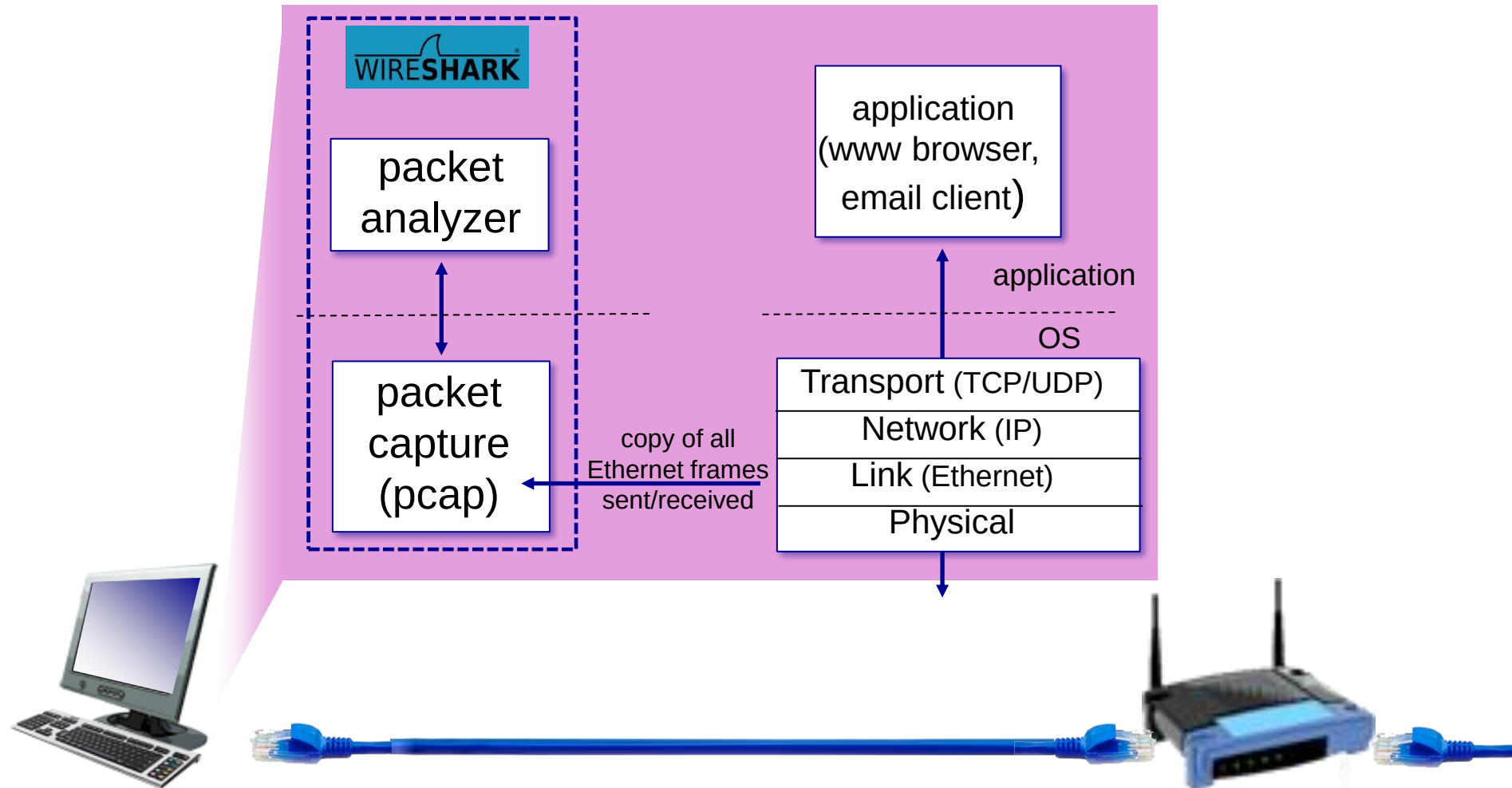
packet “sniffing”:

- broadcast media (shared Ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



Wireshark software used for our end-of-chapter labs is a (free) packet-sniffer

Wireshark

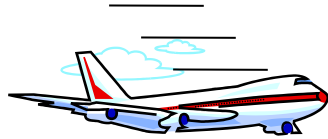


Lab 2.1: Install wireshark on your computer and do your first wireshark Lab

1. Go to <http://www.wireshark.org/download.html> and download and install the Wireshark binary on your computer.
2. Do Wireshark Lab: Getting Started v8.1
 - Download the lab instruction from Jim Kurose website
 - Follow the instructions under *Taking wireshark for a Test Run*
3. Answer Question 1-6

ภาคผนวก

Example: organization of air travel



————— *end-to-end transfer of person plus baggage* —————→

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

Example: organization of air travel



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below